

reasonable assumption depends on the particulars of the distributions being subtracted.

Assuming Independence

This is the Subtraction (Independent) Pooling Method. To subtract one distribution from another, assuming independence, on each of many iterations of a Monte Carlo procedure, BenMAP (1) randomly selects a value from the first input distribution, (2) randomly selects a value from the second input distribution, and (3) subtracts the second value from the first. To subtract N distributions from another distribution, assuming independence, BenMAP follows an analogous procedure in which, on each iteration it makes a random selection from each of the input distributions and then subtracts the second through the Nth from the first. When the Monte Carlo procedure is completed, all such generated results are sorted low to high and binned down to the appropriate number of Latin Hypercube points. The number of iterations is determined by the Monte Carlo Iterations setting.

Assuming Dependence

This is the Subtraction (Dependent) Pooling Method (see Chapter 6 for details). Recall that the uncertainty distributions in BenMAP are Latin Hypercube representations, consisting of N percentile points. To subtract one distribution from another, assuming them to be dependent, BenMAP simply generates a new N point Latin Hypercube where each point is the result of subtracting the corresponding point of the second input Latin Hypercube from the corresponding point of the first input Latin Hypercube. That is, the first point in the new Latin Hypercube is the result of subtracting the first point in the second Latin Hypercube from the first point of the first Latin Hypercube, and so forth. To subtract n distributions from another distribution, assuming dependence, BenMAP follows an analogous procedure in which each point in the new Latin Hypercube is the result of subtracting the corresponding points of the second through the Nth input Latin Hypercubes from the corresponding point of the first.

Appendix L. Command Line BenMAP

The command line version of BenMAP is capable of performing all of the functions of the GUI-based version. It is most useful for large, complex analyses that require generation of a substantial number of files. This appendix describes the syntax and use of the command line version.

L.1 Overview

The overall format of the file is a variable definitions section followed by a commands section. Comment statements are supported at any point in the file. Lines beginning with a pound character (#) are considered comment lines and will be ignored during file parsing.

Additionally, LOAD <filename> statements are supported at any point in the file. These work as string replacements - the contents of the file specified by <filename> are simply inserted into the main file. Multi-level LOAD statements are supported, but no attempt is made to detect cycles (two files referencing each other with LOAD statements, for example).

The control file is, in general, not case sensitive. In the case of user-defined strings, (variable values, etc.), it is preserved.

L.2 Variables

The variable definitions section is optional, and if present will consist of a single line with the word “Variables” on it, followed by one or more lines that define variables. A variable definition consists of a variable name and a variable value. When parsing lines in the commands section of the control file, all occurrences of the variable name will be replaced by the variable value.

All variable names must begin and end with the percent character (%).

Variable Name/Value replacement will be done in multiple passes (until no variable names remain), so variable values may contain other variable names. No attempt will be made to detect cycles, however, so be careful not to introduce them. For example, avoid variable definitions like the following:

%BENMAPDIR%	%AQGDIR%\
%AQGDIR%	%BENMAPDIR%\Air Quality Grids

Variable values must be contained in a single line, and will consist of the first non-whitespace character after the variable name through the newline character. Watch out for undesired trailing whitespaces!

L.3 Commands

The commands section is required, and will consist of one or more command sections. There are five types of command sections:

SETACTIVESETUP

CREATE AQG

RUN CFG

RUN APV

GENERATE REPORT

This section will discuss each one in turn.

In general, in command sections, there must be at least one white space between each token (where a token is either a command, a parameter name, or a parameter value).

L.3.1 Set Active Setup

For the US version of the BenMAP command line executable the only valid value is United States. The SETACTIVESETUP section is required.

Example

-ActiveSetup United States

L.3.2 Create AQG

This section initiates the creation of one or more air quality grids (normally one, potentially two in the case of monitor rollback grid creation - see below). It always starts with the words CREATE AQG. It must then include the following options, in any order:

-Filename <filename>

-Gridtype <gridtype>

-Pollutant <pollutant>

The **Filename** value is the name of the air quality grid that will be created.

The **GridType** value must be one found in the BenMAP database. The actual values for this parameter are found on the Modify Setup screen in the Grid Definitions list box.

BenMAP supports pollutant values present in the active setup. By default, these include Ozone and PM2.5 (no subscripts needed) for the United States setup. BenMAP Command Line will support additional pollutants as well, so long as they are present in

the active setup selected at the start of the Command Line Script. These values can be found in the Modify Setup screen in the Pollutants list box.

After these required options, the type of grid creation must be identified, and then the parameters for that grid creation type must be specified. There are four air quality grid creation types:

- ModelDirect
- MonitorDirect
- MonitorModelRelative
- MonitorRollback

L.3.2.1 Model Direct

This section initiates the creation of a model direct air quality grid.

This creation type has two required parameters:

- ModelFilename <filename>
- DSNName <ODBC DSN name>

and one optional parameter:

- TableName <tablename>

Supported DSNName values are:

- | | |
|--------------------|------------------------------|
| Excel Files | Excel Spreadsheet (.xls) |
| Text Files | Comma-delimited (.csv) files |
| MS Access Database | Access Database (.mdb) |

If the DSNName is “Excel Files” and there is more than one worksheet in the workbook or “MS Access Database” and there is more than one table in the database then the TableName parameter must indicate the worksheet or table name.

L.3.2.2 Monitor Direct

This section initiates the creation of a monitor direct air quality grid.

The required parameters are:

- MonitorDataType <DataSource descriptor>
- InterpolationMethod <Interpolation Method>